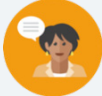


4.5 Algebraic Descriptions of a Sequence of Transformations

Lesson Introduction

 Benchmarks	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.		 Learning Targets	- I can draw an image given a preimage and an algebraic description of a sequence of transformation on a coordinate plane. - I can write a transformation on the coordinate plane using coordinate notation with algebraic descriptors.
 Benchmark Coherence	Building On MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.			N/A
 Essential Question	How can I use the information from an algebraic description of a series of transformations to draw an image?			
 Lesson Narrative	In the lesson, students continue working with drawing an image on the coordinate plane given a preimage and an algebraic description of a sequence of transformations. While students learned this in the prior lesson, students will now be doing the drawing when the sequence of transformations is presented as an algebraic description.			
 Component Summary	4.5.1 Warm-Up (~5 min.)	Write the algebraic description of a given written description (<i>SE p. 202</i>)	4.5.3 Guided Practice (10-15 min.)	Draw the results of a preimage that has gone through a sequence of transformations given as algebraic descriptions (<i>SE p. 204</i>)
	4.5.2 Guided Instruction (15 min.)	Apply a sequence of algebraic descriptions (<i>SE p. 203</i>)	4.5.4 Your Turn! (10 min.)	Draw a quadrilateral that has been transformed using a sequence of transformations presented in algebraic form (<i>SE p. 205</i>)
			4.5.5 Wrap-Up (~5 min.)	Draw a triangle on the coordinate grid that has gone through a sequence of transformations presented in algebraic form (<i>SE p. 206</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Reflection - Rotation - Translation		(Optional) Patty paper	
				Consider writing the coordinates of each transformation on the key in this guide ahead of class for ease in making quick checks to student work during the monitoring step of the lesson.

 Teacher Tips	Common Misconceptions - Students may invert the naming of shapes, use the wrong name, or believe the order does not matter. For example, students may label the 4.5.5 Wrap-Up as triangle <i>SLT</i> instead of triangle <i>TLS</i>. - Students may represent the algebraic description of a reflection of the x-axis as a change in the x-coordinate.	Things to Consider Students may choose a variety of methods for describing, writing, and drawing sequences of transformations. At this point in the unit, they could use a table of values, mapping by vertices, or graphing using descriptions. Provide space and opportunity for students to compare and contrast their methods for collective learning throughout the lesson.
	Literacy and Language Routines Compare and Connect During 4.5.2 Guided Instruction, students apply a sequence of transformations with an algebraic description. Consider a Think Aloud to make explicit connections between the visual representations and algebraic descriptions. Purposefully use language like “Compare this...” or “I am making a connection between.”	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency Students practice flexibility in their approach to sequences of transformations, using a variety of methods for drawing images using written descriptions, tables of values, or algebraic descriptions. The clarifications in this standard include allowing students to select a procedure, with “efficiency” meaning different things for different students based on their learning entry points.
	 Differentiate Instruction For a different entry point, consider first breaking the sequence of transformations into three separate problems and having the students draw each separate transformation on separate same-sized grids. Then have the students overlay each transformation to copy all three transformations onto one grid. Making this move will help students who may become overwhelmed by the amount of drawing in one space.	
Lesson Notes:		

4.5.1 Warm-Up: Algebraic Descriptions of Transformations

Component Overview: Students will complete a table where they are writing the algebraic description of a given written description. For this activity, allow students to refer to their previously written notes. Encourage students to refine any mistakes they may discover in their notes.



4.5.1 Warm-Up: Algebraic Descriptions of Transformations

- Use your notes to complete the algebraic descriptions in the table.

The Effect of Transformations on Coordinates

a.

Rotations Clockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the x -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y -coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (y, -x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (-y, x)$
Rotations Counterclockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the y -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x -coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (-y, x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (y, -x)$

b.

Translations	
Right	Left
Add to the x -coordinate	Subtract from the x -coordinate
$(x, y) \rightarrow (x + a, y)$	$(x, y) \rightarrow (x - a, y)$
Up	Down
Add to the y -coordinate	Subtract from the y -coordinate
$(x, y) \rightarrow (x, y + b)$	$(x, y) \rightarrow (x, y - b)$

c.

Reflections	
x -axis	y -axis
The y -coordinate changes to its opposite.	The x -coordinate changes to its opposite.
$(x, y) \rightarrow (x, -y)$	$(x, y) \rightarrow (-x, y)$
$y = x$	$y = -x$
The coordinates switch.	The coordinates switch and become the opposite.
$(x, y) \rightarrow (y, x)$	$(x, y) \rightarrow (-y, -x)$



Notice the similarities and differences between this activity and the summary found in the Guided Instruction in Lesson 4. Consider having students use their notes from that activity or from the Bring It Together activities found in Lessons 1 and 2 as a verification step when completing the Warm-Up.

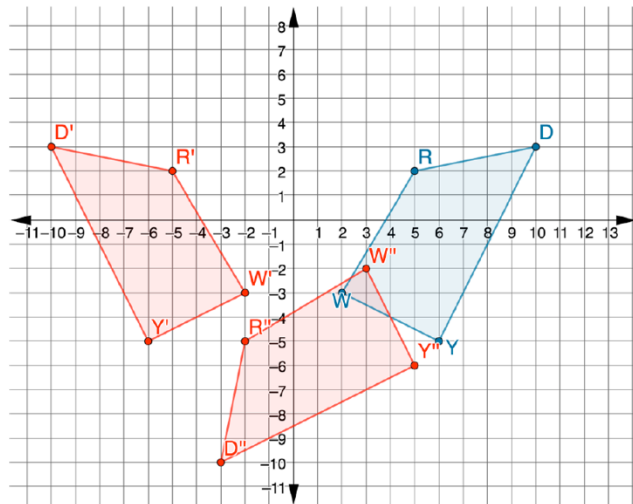
4.5.2 Guided Instruction: Applying Algebraic Descriptions

Component Overview: Students will receive guided instruction on applying a sequence of transformations written as algebraic descriptions. For this activity, leverage the “I Do” and “We Do” components of the gradual release model.



4.5.2 Guided Instruction: Applying Algebraic Descriptions

1. Quadrilateral $RDYW$ is shown on the coordinate grid.



Apply the sequence of transformations to quadrilateral $RDYW$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \rightarrow (-x, y)$ Label as $R'D'Y'W'$.
- $(x, y) \rightarrow (-y, x)$ Label as $R''D''Y''W''$.

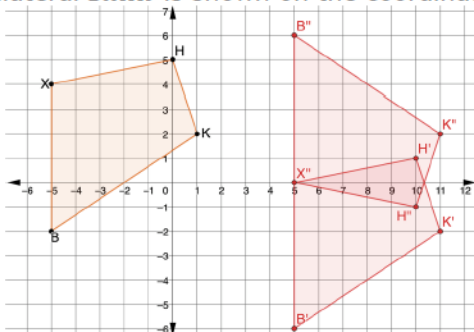
Shown on graph.



If available, consider the use of physical manipulatives or patty paper to help students see each step of the transformation occur as it happens during the sequence.

4.5.2 Guided Instruction: Applying Algebraic Descriptions (continued)

2. Quadrilateral $BXHK$ is shown on the coordinate grid.

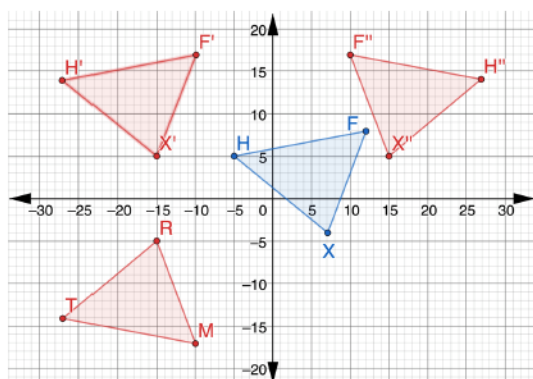


Apply the series of transformations to quadrilateral $BXHK$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \rightarrow (x + 10, y - 4)$ Label as $B'X'H'K'$.
- $(x, y) \rightarrow (x, -y)$ Label as $B''X''H''K''$.

Shown on graph.

3. Triangle HFX is shown on the coordinate grid.



Perform the sequence of transformations on $\triangle HFX$. Assume all rotations are centered at the origin. Label the result of all three transformations as $\triangle TMR$.

- $(x, y) \rightarrow (x - 22, y + 9)$
- $(x, y) \rightarrow (-x, y)$
- $(x, y) \rightarrow (-x, -y)$

Shown on graph.



Consider using some or all of the following prompts for inquiry-based learning during this activity:

- What transformation is being applied when you see $(x + 10, y - 4)$? How can you tell? What specifically is asked to be done?
- How would the table of values of the vertices of $BXHK$ be changed when applying $(x + 10, y - 4)$?
- How could you show the transformation of the Vertex K using mapping?
- What transformation is being applied when you see $(x, -y)$? How can you tell? What specifically is asked to be done?
- How would the table of values of the vertices of $B'X'H'K'$ be changed when applying $(x, -y)$?
- How could you show the transformation of the vertex K' using mapping?



When reviewing question 2, consider having the students create a table of values of the vertices of triangle HFX and complete a new table of values for each transformation. Then support students who prefer mapping by showing the mapping of one vertex.



If time permits, allow students to share their answers for question 3 with a partner before showing the class the answer. This will allow students the opportunity to revise their thinking after discussing with their peers before the answers are shared whole group.

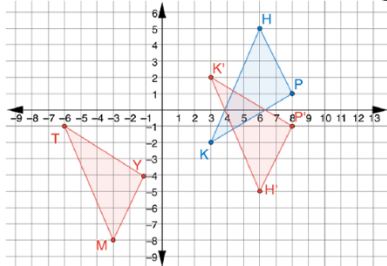
4.5.3 Guided Practice: Applying Algebraic Descriptions

Component Overview: Students will practice with drawing images that are the results of a preimage that has gone through a series of transformations. For this activity, allow students to try on their own and then compare their work with a partner.



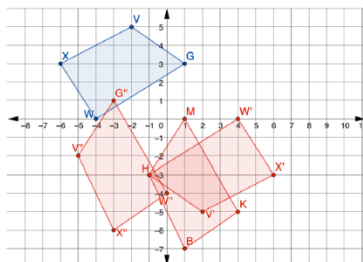
4.5.3 Guided Practice: Applying Algebraic Descriptions

1. Triangle PKH is shown on the coordinate grid.



- a. Perform the sequence of transformations on $\triangle PKH$. Assume all rotations are centered at the origin. Label the result of transformations as $\triangle YTM$.
- $(x, y) \rightarrow (x, -y)$
 - $(x, y) \rightarrow (x - 9, y - 3)$
- Shown on graph.*
- b. Check your answer with your partner.

2. Quadrilateral $XVGW$ is shown on the coordinate grid.



Perform the series of transformations on quadrilateral $XVGW$. Assume all rotations are centered at the origin. Label the result of transformations as quadrilateral $BHMK$.

- $(x, y) \rightarrow (-x, -y)$
- $(x, y) \rightarrow (y, -x)$
- $(x, y) \rightarrow (x + 4, y - 1)$

Shown on graph.



While the primary intent of this activity is an individual activity with a partner review, consider allowing students to work with a partner from the beginning of the Guided Practice. If you do this option for question 1, release students to work on question 2 on their own and then check their answers with their partner.



- Did you and your partner both choose the same method for determining the location of $\triangle YTM$?
- Did you and your partner arrive at the same answer initially? If not, what did you notice that was different?
- Were you able to agree on the final answer?



Consider strategies for breaking down the sequence into three parts representing each step of the sequence or each separate transformation.



Note that three transformations are used in this sequence.



For students who finish early, consider having students create a fourth transformation, displayed algebraically, and add that fourth transformation to their coordinate grid.

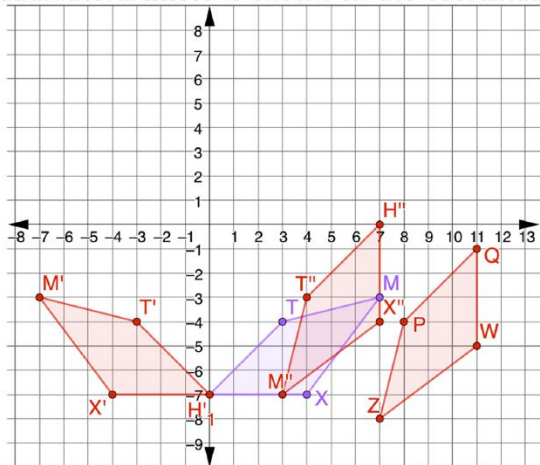
4.5.4 Your Turn!

Component Overview: Students will independently practice drawing a quadrilateral using a sequence of transformations presented in algebraic form.



4.5.4 Your Turn!

1. Quadrilateral $XHTM$ is shown on the coordinate grid.



Perform the sequence of transformations on quadrilateral $XHTM$. Assume all rotations are centered at the origin. Label the result of the transformations as quadrilateral $WQPZ$.

- $(x, y) \rightarrow (-x, y)$
- $(x, y) \rightarrow (-y, x)$
- $(x, y) \rightarrow (x + 4, y - 1)$

Shown on graph.



This activity can be modified for students who may need a challenge or who may need additional scaffolded support.

- *Extension:* Determine if the transformations could be done in the reverse order.
- *Scaffolded:* Reduce the number of transformations down to two.



- *What are the types of transformations being asked for in this question?*
- *How can you tell based on the algebraic description?*

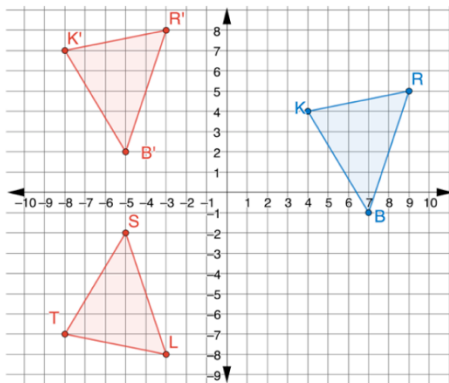
4.5.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that has students draw a triangle on the coordinate grid that has gone through a sequence of transformations presented in algebraic form. This activity should be done independently to determine student mastery of this lesson's content.



4.5.5 Wrap-Up: Ticket Out the Door

- Triangle KRB is shown on the coordinate grid.



Consider prompting students to provide written descriptions for each transformation, as they may properly identify the type of transformation but still struggle mapping its result.

Perform the sequence of transformations on $\triangle KRB$. Assume all rotations are centered at the origin. Label the result of the transformations as $\triangle TLS$.

- $(x, y) \rightarrow (x - 12, y + 3)$
- $(x, y) \rightarrow (x, -y)$

Shown on graph.